

Course Notes: Lecture 9 & Multi-Parameter Models

CLAD IV

August 2026

Lecture 9: Overview & Administration

Lecture Details & Quiz Logistics

- **Date/Time:** Tuesday, August 10, 2026, 2:32 PM
- **Course:** CLAD IV
- **Quiz Schedule:** 40 mins quiz on 24th August, 4:00 PM – 4:40 PM
- **Weightage:** 5% of the course
- **Policy Note:** (−25/100) for cheating (if you're fined, marks at the end of the course are safe). Then 25/100.

Fisher Information & Prior Derivations

- Loss / Function relations: $G(\cdot)$, $L\left(\frac{\theta}{\hat{\theta}}\right)$
- $\Phi_i = -f_i(\theta)$
- Prior proportionality: $\pi_T(\theta) \propto \sqrt{I(\theta)}$
- Information identity: $E\left(\frac{\partial^2 \log L}{\partial \theta^2}\right) = -I(\theta)$
- Reparameterization ($\phi = \text{fn}(\theta)$):

$$I(\phi) = I(\theta) \left(\frac{d\theta}{d\phi}\right)^2$$

- Variance bound: $\text{Var}(T(x)) \geq \frac{1}{I(\theta)}$
- Variance and distributions: $\text{Var}(\cdot)$, $\text{Poisson}(\cdot)$, $\frac{k(1-k)}{n}$

Jeffreys' Prior Derivations

Poisson Distribution:

Derive Jeffreys' prior: $\pi_J(\lambda) \propto \frac{1}{\sqrt{\lambda}}$

$$\int_k^{\mathbb{R}} \pi_T(x) |dx|, \quad k_L(\theta_\lambda, k_\lambda d\lambda) = k_L(\pi ds)^2 dt$$

Binomial Distribution:

Derive for Binomial: $\theta = \frac{p}{1-p} \implies \phi = \log \frac{p}{1-p}$

Miscellaneous Notes & Vector Representations

- Matrix / Column Vector structure:

$$\begin{pmatrix} A_x \\ X_y \\ X_y \\ X_y \\ X_y \\ X_y \\ X_z \\ Y_z \end{pmatrix}$$

- Differential relation: $(x, \Omega) d\lambda = 2\sqrt{3}$
- $F \rightarrow (x, x^2)$ Brown $\int \frac{P}{\pi_T}(p) d^2 P$
- $\pi(\theta) = \text{Beta}(\frac{1}{2}, \frac{1}{2})$
- Transformation: $x = z + (1-x)$ where $z \sim N(0, s^2)$
- $\pi_1(\nu) = \frac{e}{\text{Jeffreys}}(\frac{1}{2}, \nu)$ NIU $x = z + \boxed{\rho 1}$
- $\pi_\xi(\cdot) \quad \left(-1(\sqrt{n^2}, e^2)\right)$

Priors Taxonomy

- Non-informative / Jeffreys prior
- Flat prior: $\pi(\theta) \propto 1$
- Diffuse prior: $(\pi_0(\cdot))$
- Normal mean known, unknown σ^2 : $\pi(\sigma^2) \propto \frac{1}{\sigma^2}$
- Multi-parameter models: $2 \sin^{-1}(\sqrt{p}), \quad [b_1, m_1, p_2]$

Multi-Parameter Models

Wednesday, August 19, 2026 – 2:05 PM

1. Setup and Nuisance Parameters

- Prior distribution over parameters: $\pi(\theta_1, \theta_2)$
- Data distribution: $X_1, \dots, X_n \mid \theta_1, \theta_2 \sim f(x; \theta_1, \theta_2)$
- Example (Normal Model): $X_1, \dots, X_n \mid \mu, \sigma^2 \sim N(\mu, \sigma^2)$
- Joint posterior distribution:

$$\pi(\theta_1, \theta_2 \mid D) \propto \pi(\theta_1, \theta_2) \pi(D \mid \theta_1, \theta_2)$$

- **Nuisance Parameters (Marginalization):**

$$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D) d\theta_2$$

$$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D) d\theta_2 = \int \pi(\theta_1 \mid \theta_2, D) \pi(\theta_2 \mid D) d\theta_2$$

Monte Carlo Integration & Sampling

Sampling sequence: $(X^{(1)}, Y^{(1)}), (X^{(2)}, Y^{(2)}), \dots$ where $\pi(x, y) = \pi(x \mid y)\pi(y)$.

- Samples generated: $\theta_1^{(g)}, \theta_2^{(g)}$
- The marginal distribution of θ_1 can be obtained simply by ignoring the samples of θ_2 .
- $\pi(\theta_2 \mid D) : \theta_2 \rightarrow (M_1), M_2 \implies \pi(\theta_1 \mid M_1, D)$ (Single-parameter inference).

2. Estimation in $N(\mu, \sigma^2)$ Model

Suppose you want to estimate μ & σ^2 in $N(\mu, \sigma^2)$ model. For a known value of σ^2 , you know how to estimate, and you also know $p(\sigma^2 \mid D)$.

$$\pi(\mu \mid D) = \int \pi(\mu \mid \sigma^2, D) \pi(\sigma^2 \mid D) d\sigma^2 \quad (\text{to be calculated})$$

Sampling Algorithms

Generating samples from $\pi(\mu \mid D)$ in 2 steps:

Step 1: Generate samples from $\pi(\sigma^2 \mid D)$ (call it $\sigma^{2(g)}$).

Step 2: Generate sample from conditional distribution $\pi(\mu \mid \sigma^{2(g)}, D) \rightarrow \mu^{(g)}$.

Gibbs Sampling (Conditional):

- $\sigma^2 \mid \mu, D \rightarrow \sigma^{2(g)}$
- $\mu^{(g+1)} \mid \sigma^{2(g+1)}, D \rightarrow \mu^{(g+1)}$
- Go to Step 1 & Iterate.

3. Scaled-Inverse- χ^2 Distribution

Density function:

$$f(x; \nu, s^2) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \frac{(s^2)^{\nu/2}}{x^{(\nu/2+1)}} e^{-\frac{\nu s^2}{2x}}, \quad x > 0$$

where ν represents degrees of freedom, and s^2 is the scale parameter.

Properties of Scaled-Inv- $\chi^2(\nu)$

If $X \sim \text{Scaled-inv-}\chi^2(\nu, s^2)$, then:

- a) $\frac{\nu s^2}{X} \sim \chi^2(\nu)$
- b) $X \sim \text{Inv-Gamma}\left(\frac{\nu}{2}, \frac{\nu s^2}{2}\right)$
- c) $kX \sim \text{Scaled-inv-}\chi^2(\nu, ks^2)$

Proof of (c):

$$\frac{\nu s^2 k}{kX} \sim \chi^2(\nu) \implies kX \sim \text{Scaled-inv-}\chi^2(\nu, ks^2)$$

4. Conjugate Inference & Predictive Distributions

Normal model with unknown σ^2 , known μ :

- Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, where μ is known.
- Prior: $\sigma^2 \sim \text{Scaled-inv-}\chi^2(\nu_0, \sigma_0^2)$
- Posterior: $\sigma^2 \mid D \sim \text{Scaled-inv-}\chi^2(\nu_n, s_n^2)$

$$\text{where } \nu_n = \nu_0 + n, \quad s_n^2 = \frac{\nu_0 \sigma_0^2 + \sum_{i=1}^n (X_i - \mu)^2}{\nu_0 + n}$$

Posterior Predictive Distribution

Let $Y \sim N(\mu, \sigma^2)$ where μ is known. Find the marginal distribution $f(y \mid D)$:

$$\frac{Y - \mu}{s_n} \sim t_{\nu_n} \quad (\text{DO IT})$$

Integration formula:

$$f(y \mid D) = \int f(y \mid \sigma^2) \pi(\sigma^2 \mid D) d\sigma^2$$

Since $Y \mid \sigma^2 \sim N(\mu, \sigma^2)$ and $\sigma^2 \mid D \sim \text{Scaled-inv-}\chi^2(\nu_n, s_n^2)$, the marginal distribution of Y is Student- t ($t_{\nu_n}(\mu, \sigma_0^2)$).

5. Unsolved / Exercise Problem

μ unknown & σ^2 follows non-informative prior:

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2} \longrightarrow \text{Find } (\sigma^2 \mid y)?$$